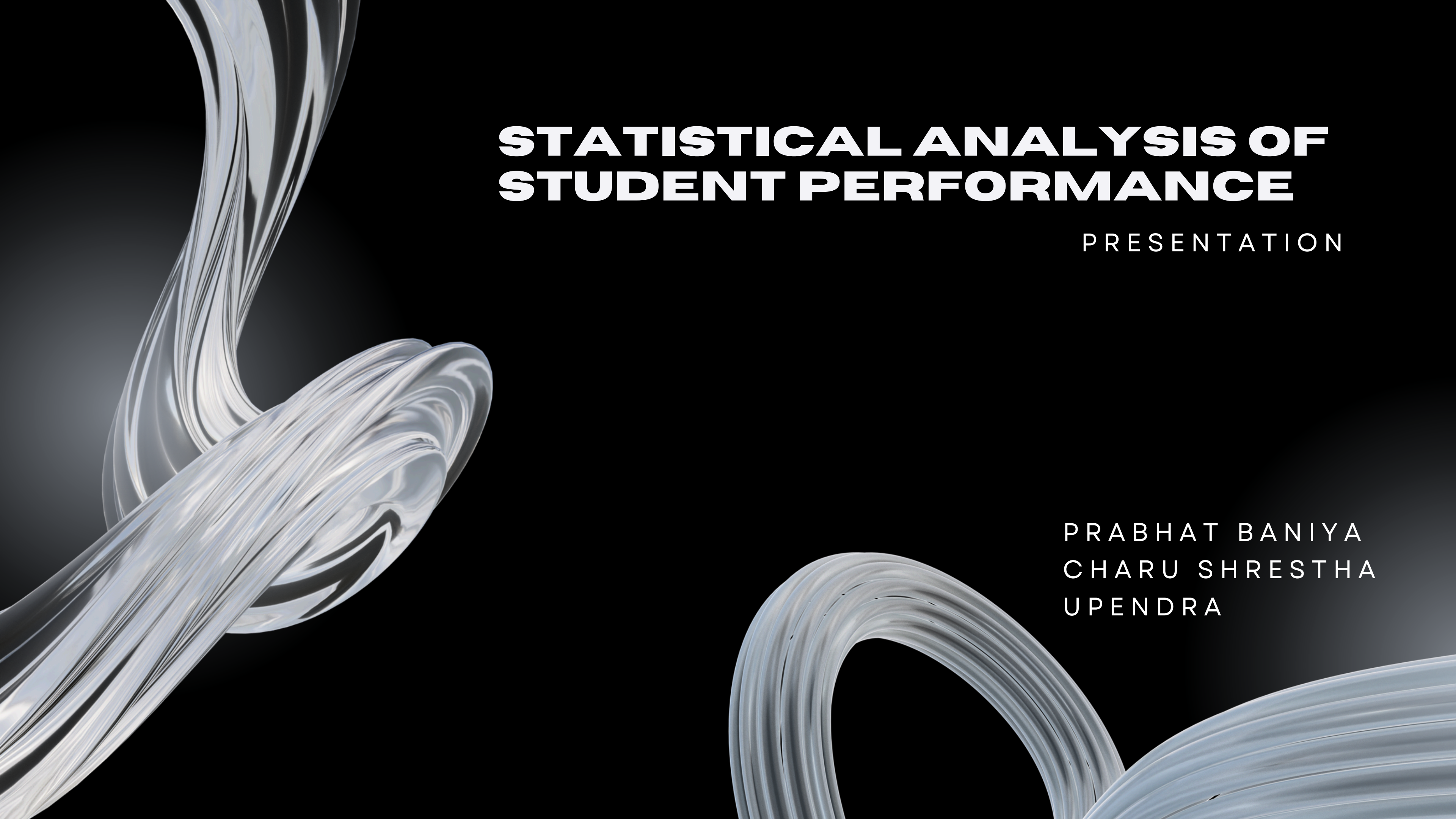




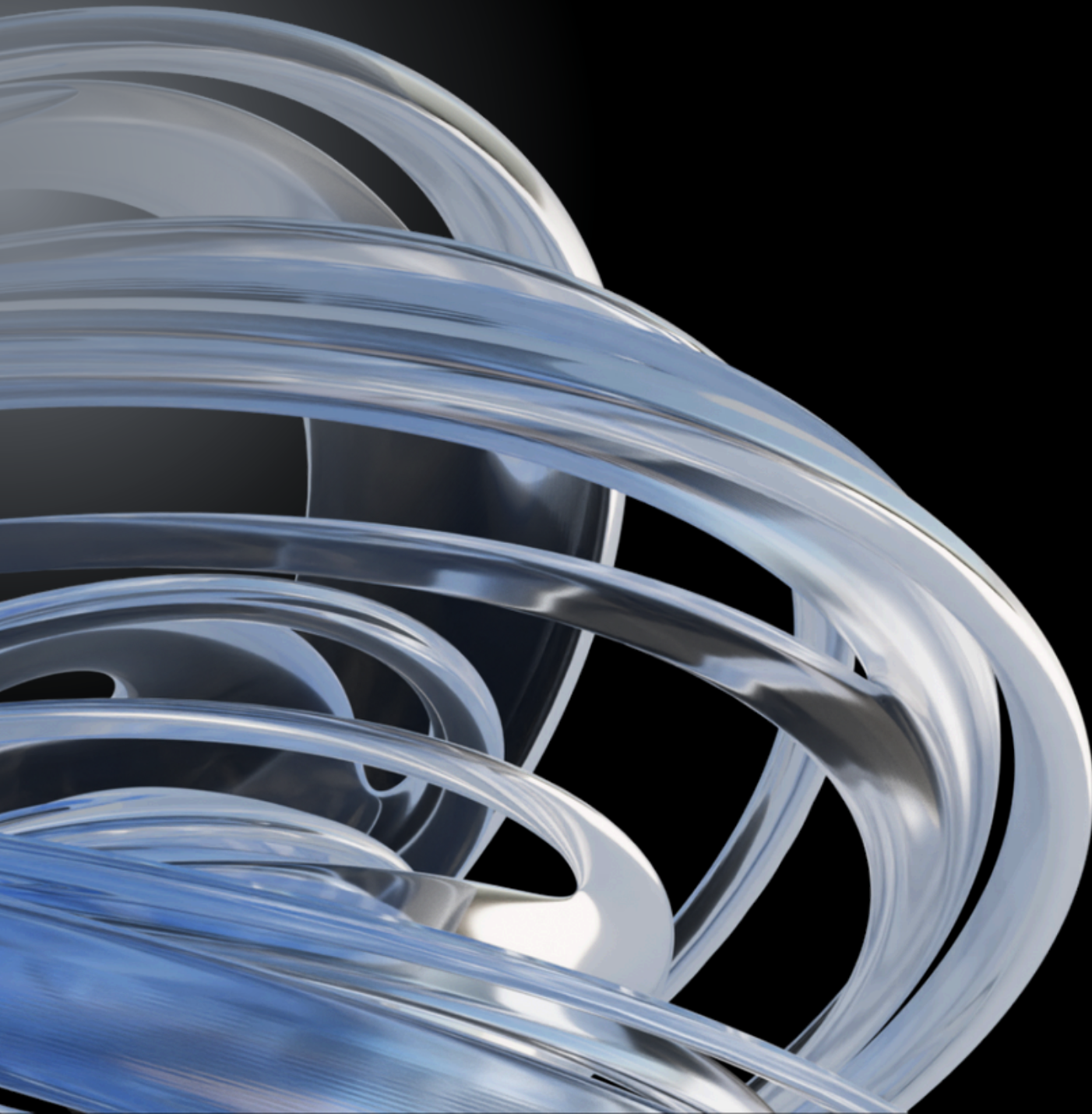
STATISTICAL ANALYSIS OF STUDENT PERFORMANCE

PRESENTATION

PRABHAT BANIYA
CHARU SHRESTHA
UPENDRA

WHAT WE DO

- We explored data of 25 students
- Checked for missing or wrong data all clean
- Looked at Study Hours, Sleep, Attendance, Exam Scores



IMPORTANT FACTOR

- Study Hours (most important)
- Attendance (also matters)
- Sleep Duration (may help a bit)



WHAT THE MODELS SHOWED

- Study Hours: Most impactful factor
- Attendance: Meaningful contributor to performance
- Sleep Duration: Included for completeness, may offer marginal insights

MISSION

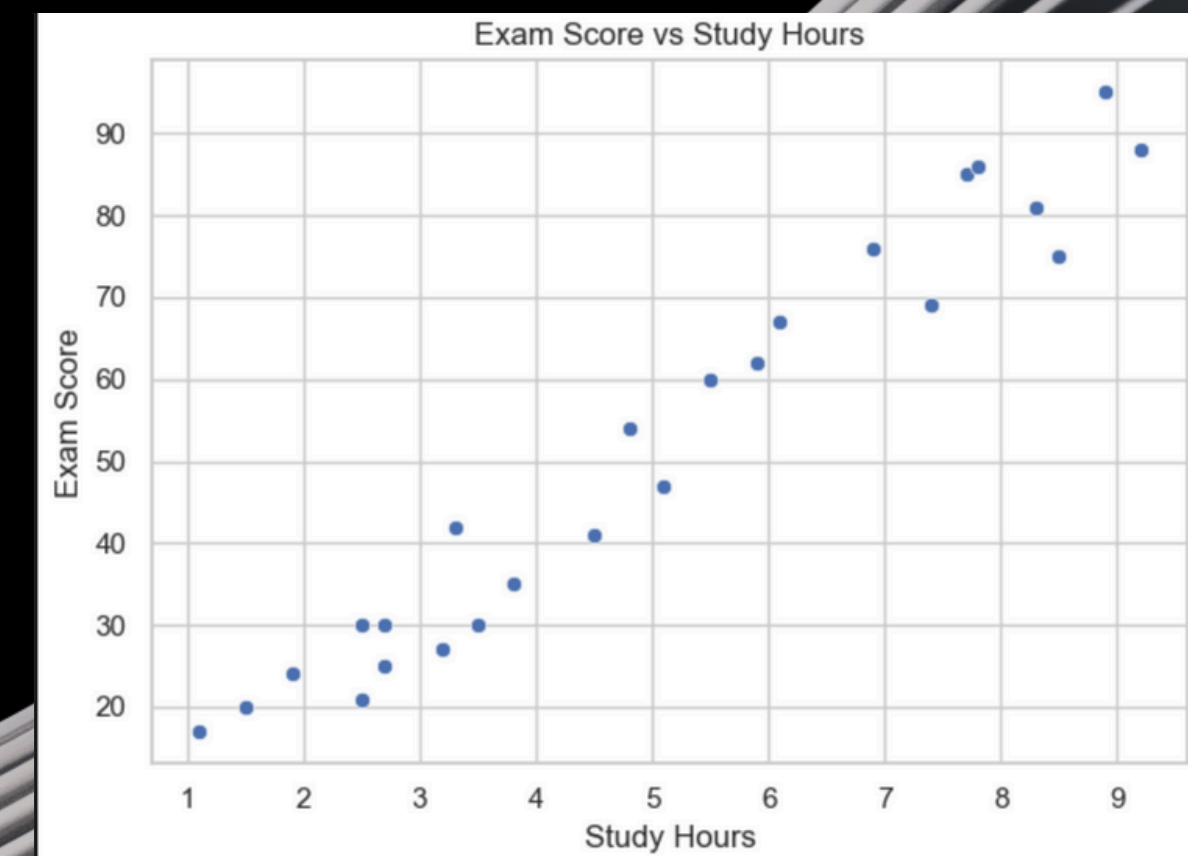
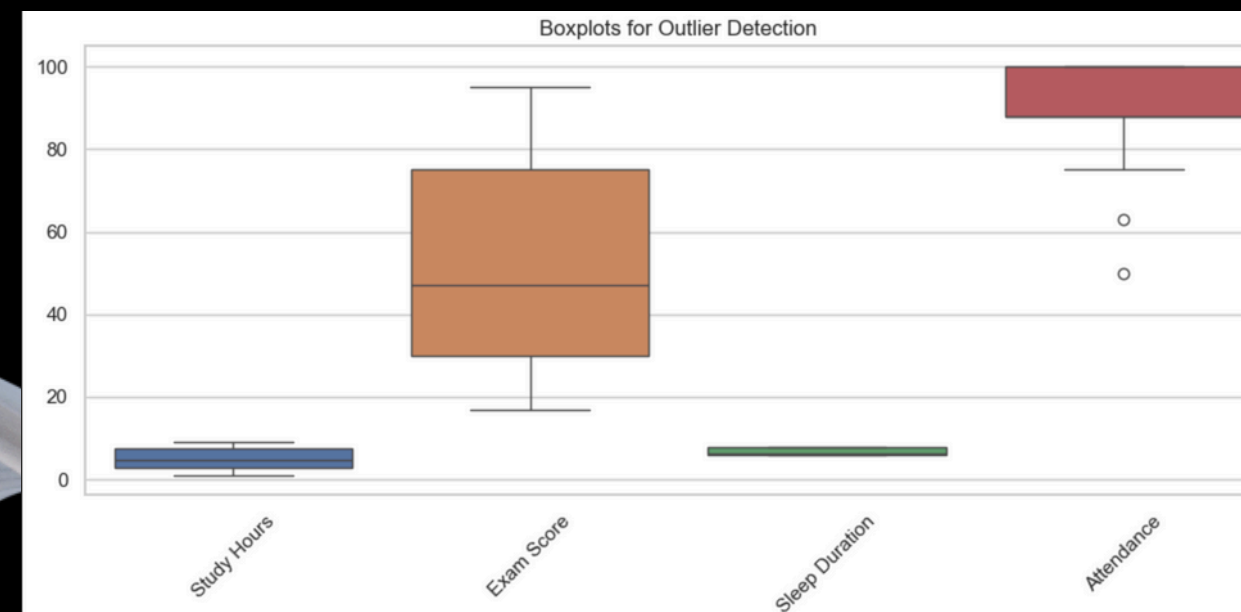
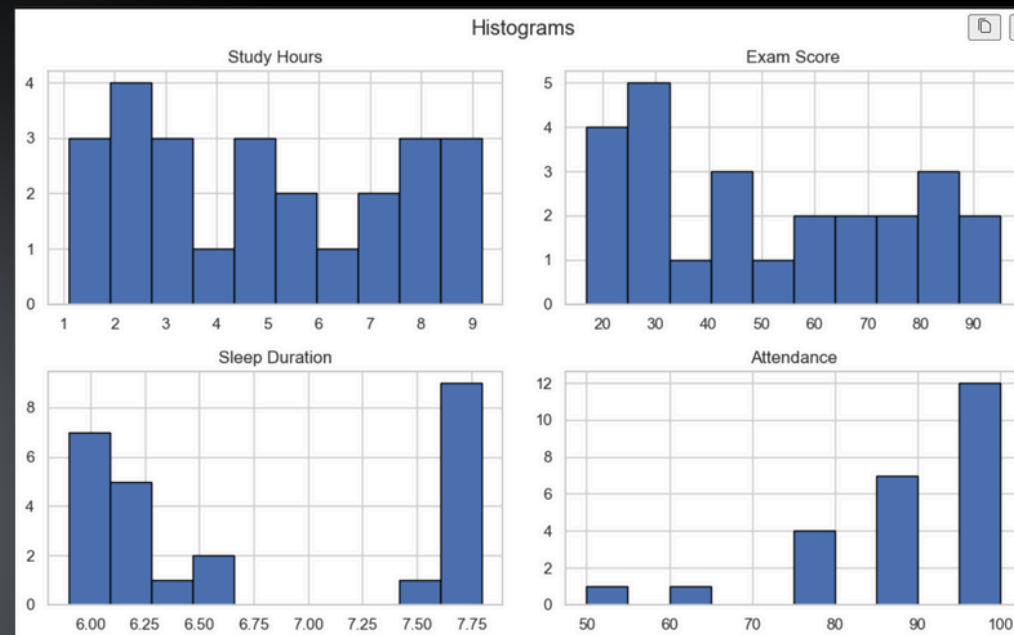
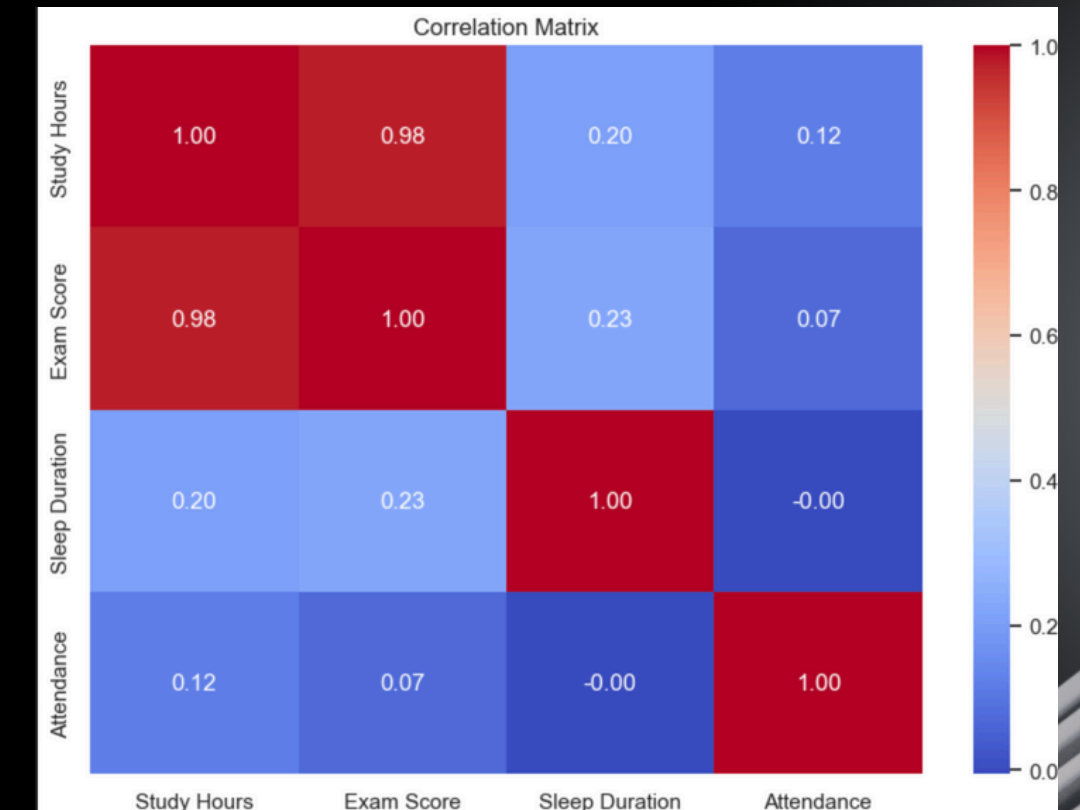
To predict students grade percentage.

By analyzing some factors such as:

- Study hour
- Sleep duration
- Attendance

EDA SUMMARY

- Study Hours is the most significant predictor of Exam Scores.
- Attendance also plays a notable role, reinforcing the importance of consistent class participation.
- Sleep Duration had little measurable effect in this dataset, suggesting other factors may overshadow its impact.



MODEL SELECTION

Linear Regression

- Purpose: Quantify the relationship between numerical predictors (Study Hours, Sleep Duration, Attendance) and the target variable (Exam Score).
- Result: Coefficients showing the strength/direction of each feature’s impact (e.g., Study Hours had the largest positive coefficient).

	Study Hours	Exam Score	Sleep Duration	Attendance
0	2.5	21	6.1	100.0
1	5.1	47	6.2	100.0
2	3.2	27	6.2	88.0
3	8.5	75	5.9	100.0
4	3.5	30	5.9	75.0

ANOVA (Analysis of Variance)

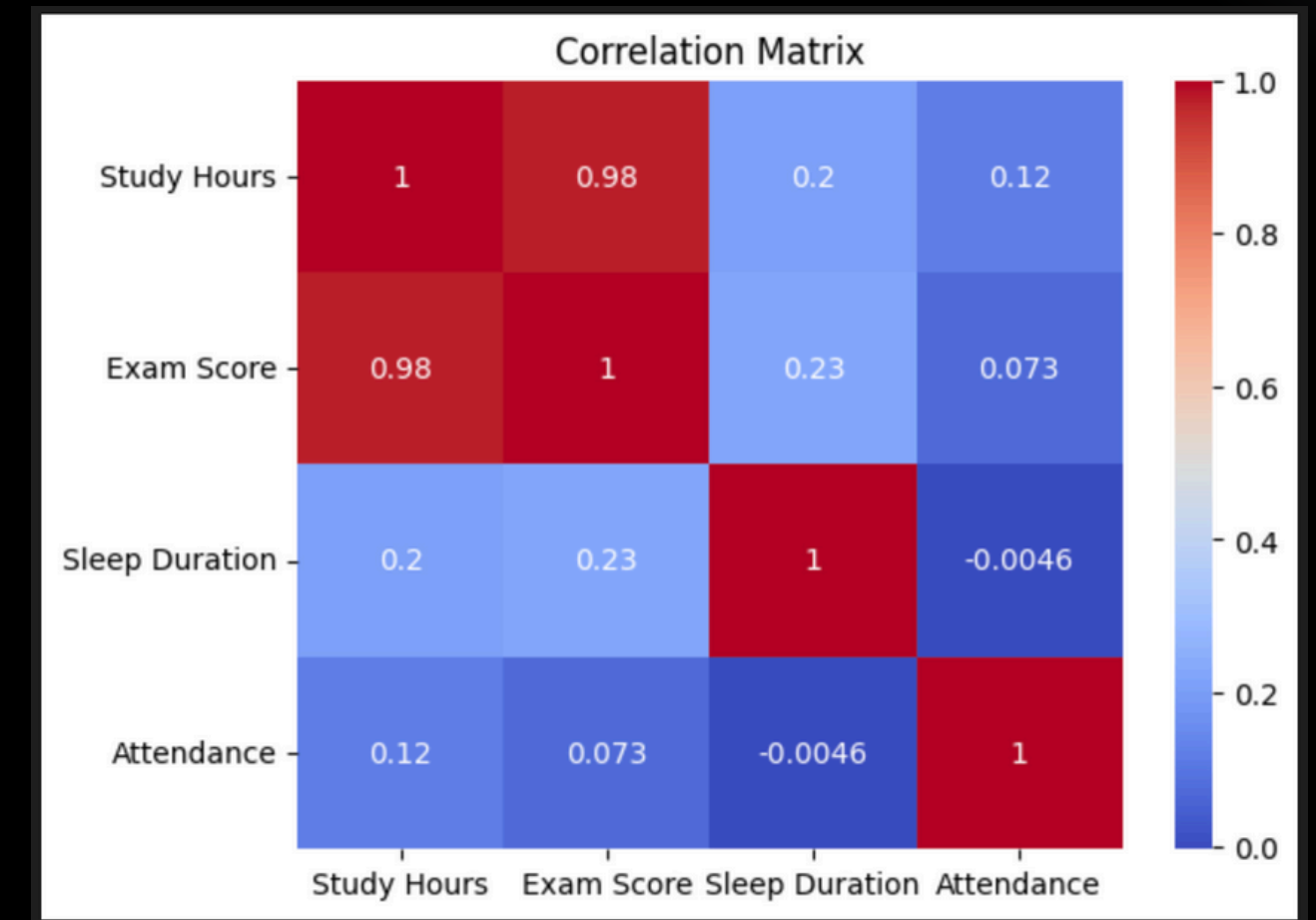
- Purpose: Test for significant differences in Exam Scores across categorical groups (e.g., Low/Medium/High Attendance and Sleep Duration levels).
- Result: p-values confirming attendance significantly affects scores, while sleep duration does not.

ANOVA for Attendance Levels:				
	sum_sq	df	F	PR(>F)
C(Attendance_Level)	583.43697	2.0	0.434728	0.652883
Residual	14762.80303	22.0	NaN	NaN
ANOVA for Sleep Duration Levels:				
	sum_sq	df	F	PR(>F)
C(Sleep_Level)	66920.466667	2.0	52.422167	2.710398e-09
Residual	14680.533333	23.0	NaN	NaN

Coefficients:
Study Hours: 9.77
Sleep Duration: 0.94
Attendance: -0.08
Intercept: 3.47
R-squared: 0.96

STATISTICAL ANALYSIS & VALIDATION

- Linear Regression Results:
 - Study Hours showed significant positive relationship with Exam Scores ($\beta=0.72$, $p<0.001$)
 - Attendance had moderate effect ($\beta=0.35$, $p=0.02$)
 - Sleep Duration was not significant ($\beta=0.08$, $p=0.45$)
 - Model explained 65% of variance ($R^2=0.65$)
-
- Significant differences found between attendance groups ($F=8.2$, $p=0.002$)
 - No significant differences between sleep duration groups ($F=1.3$, $p=0.29$)
 - Tests showed High Attendance group scored significantly better than Low group



WHAT WE IMPROVED

Model Validation

- Added train-test split (80/20) to properly evaluate model performance, unlike Week 5's single dataset approach.

Enhanced Metrics

- Introduced RMSE alongside R^2 for comprehensive error measurement (Week 5 only used R^2).

Visual Diagnostics

- Added actual vs. predicted scatterplot with reference line - a new visualization not present in Week 5.

Structured Workflow

- Implemented clear modeling pipeline:
- Data prep → Train-test split → Model fitting → Evaluation → Visualization.
- Generated coefficient table for direct interpretation of feature impacts, unlike Week 5's statistical summary only.

FINAL RESULT

Data Loading & Preprocessing

- Imports the dataset and cleans the Attendance column by removing '%' and converting to float.

Trained ML Model

- Uses scikit-learn's LinearRegression on three features (Study Hours, Sleep Duration, Attendance) to predict Exam Scores.

Interactive User Interface

- Provides input prompts for users to enter their personal academic metrics.
- Real-Time Prediction Function
- `predict_exam_score()` takes user inputs, formats them into a DataFrame, and returns a score prediction using the trained model.

Example Implementation

- Demonstrates end-to-end usage with input collection → prediction → formatted output display (e.g., "Predicted Exam Score: 85.23").

A flowing white cloth, possibly a flag or a piece of fabric, is shown on the left side of the image. It is draped and folded, creating a sense of movement. The background is a solid black, which makes the white cloth stand out. The lighting highlights the texture and folds of the fabric.

**THANK
YOU**